

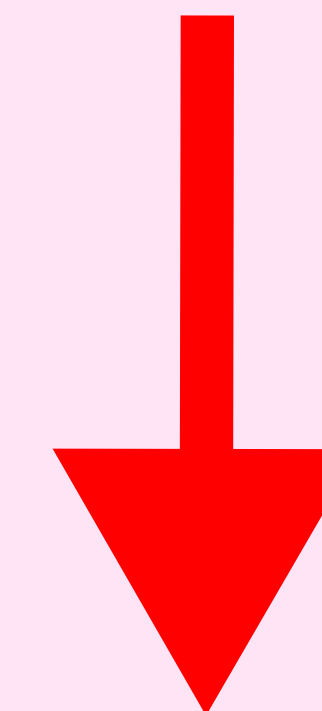
Why is this a hard problem?

Why not replace these with hand-tuned libs?

```
FOR I = 1, N:  
  Arr[I] = Arr[I]+1  
  FOR J = i, N:  
    FOR K = 1, N:  
      // other code  
      C[I, J] += A[I, K] * B[K, J]  
      T[i] = C[I, J]  
    ENDFOR  
    // more code  
  ENDFOR  
ENDFOR
```

The major challenge is extracting hidden (latent) representation, which programmers can't do manually

Extract domain specific parts from general purpose programs



Then use the domain specific compiler to optimize those parts!